Observer Quality as a Resource Constraint in Quantum Darwinism: Optimal Decoding, ε-Approximate Spectrum Broadcast Structure, and a Central-Spin Worked Example

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Quantum Darwinism (QD) and Spectrum Broadcast Structure (SBS) formalize how many observers can infer the same pointer value of a system by accessing different fragments of its environment. Most QD analyses treat observers as ideal. We introduce an explicit observer-quality triple $Q_O = (R_O, \Lambda_O, \tau_O)$ encoding access fraction, calibration noise (as a CPTP map), and temporal integration horizon. Replacing conditional i.i.d. record assumptions with an explicit ε-SBS trace-distance error model, we propagate observer constraints through redundancy/objectivity statements using decoder-level continuity. Our main results are: (i) Chernoff-optimal sample-complexity bounds for the number of fragments required to infer the pointer value under calibration, (ii) a data-processing theorem showing calibration noise degrades the quantum Chernoff exponent, and (iii) ε-robust upgrades from ideal SBS to approximate SBS. We provide a worked open-system example (central-spin pure dephasing) computing observer parameters from Hamiltonian couplings and readout noise, and we prove a sharp exponent gap between collective and product measurements for pure-state records, directly linking temporal observer constraints to redundancy.

I. INTRODUCTION

Quantum Darwinism (QD) proposes that classical objectivity arises when information about a pointer observable is redundantly encoded in the environment and can be accessed by many observers [1–3]. Spectrum Broadcast Structure (SBS) provides a structural characterization of objective states [4, 5], and strong-QD connects entropic and structural conditions [5, 6].

A recurring modeling gap is that the observer is typically idealized. Yet in realistic settings observers have limited access, imperfect calibration, and finite temporal horizons. While there is substantial prior work on non-ideal *environments* (e.g., hazy environments [7]) and on generic emergence of objectivity [8, 9], explicit observer parameterizations remain sparse. Operational approaches emphasize *accessible* information in fragments rather than mutual information [10].

We focus on one task: derive observer-relative redundancy/sample-complexity statements that remain stable under an explicit ε-SBS trace-distance approximation. Technically, we recast "learning the pointer value" as a hypothesis testing problem and use the quantum Chernoff bound [11, 12] to obtain tight (exponent-optimal) scaling.

II. SETUP: SBS, ε-SBS, AND OBSERVER QUALITY

Let S be the system and $\mathcal{E} = \mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_N$ an environment partition. We focus on a pointer variable X taking values in a finite alphabet $\mathcal{X}$ (often $\mathcal{X} = \{0, 1\}$).

Definition 1 (Spectrum Broadcast Structure (SBS))**.** A state $\sigma_{S\mathcal{E}}$ has *SBS* with respect to pointer basis $\{|x\rangle\}_{x\in\mathcal{X}}$ if

$$\sigma_{S\mathcal{E}} = \sum_{x\in\mathcal{X}} p_x \, |x\rangle\langle x|_S \otimes \bigotimes_{k=1}^{N} \sigma_{\mathcal{E}_k}^{(x)}, \tag{1}$$

where, for each k, the conditional states $\{\sigma_{\mathcal{E}_k}^{(x)}\}_{x\in\mathcal{X}}$ have mutually orthogonal supports (perfect distinguishability) [4, 5].

Definition 2 (ε-SBS)**.** A state $\rho_{S\mathcal{E}}$ is *ε-SBS* if there exists an SBS state $\sigma_{S\mathcal{E}}$ such that $D_{\mathrm{tr}}(\rho_{S\mathcal{E}}, \sigma_{S\mathcal{E}}) \leq \varepsilon$, where $D_{\mathrm{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$.

Definition 3 (Observer quality)**.** An observer is specified by a quality triple $Q_O = (R_O, \Lambda_O, \tau_O)$: R_O is an access fraction (at most $m_{\max} = \lfloor R_O N \rfloor$ fragments per episode), Λ_O is a CPTP calibration/preprocessing map applied to each accessed fragment, and τ_O is a temporal horizon (acquisition/memory time).

When $\mathcal{X} = \{0, 1\}$, we also define a scalar *single-fragment calibration index* from the Helstrom success probability.

Definition 4 (Binary calibration index)**.** Let ρ_0, ρ_1 be the two conditional states of a single accessed fragment after Λ_O, with equal priors. Define

$$C_O := 1 - P_e^\star(\rho_0, \rho_1), \qquad P_e^\star(\rho_0, \rho_1) = \tfrac{1}{2}\left(1 - \tfrac{1}{2}\|\rho_0 - \rho_1\|_1\right), \tag{2}$$

where $P_e^\star$ is the Helstrom-optimal Bayes error [13, 14].

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III. CHERNOFF DECODING BOUNDS AND CALIBRATION CONTRACTION

For a binary pointer $X \in \{0,1\}$ and a fragment set $A \subseteq \{1,\dots,N\}$, define the conditional states of the accessed environment under an SBS witness experiment: $\sigma_A^{(x)} := \bigotimes_{k\in A} \sigma_{\mathcal{E}_k}^{(x)}$. The observer applies Λ_O to each accessed fragment and then performs an optimal measurement to decide x. This yields the hypothesis testing problem $H_0 : \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})$ versus $H_1 : \bigotimes_{k\in A} \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})$.

Define the quantum Chernoff coefficient $Q(\rho,\sigma) = \min_{s\in[0,1]} \mathrm{Tr}[\rho^s \sigma^{1-s}]$ and exponent $\xi(\rho,\sigma) = -\log Q(\rho,\sigma)$ [11, 12].

Theorem 1 (Finite-n Chernoff upper bound)**.** *For equal priors on $H_0 : \rho$ vs $H_1 : \sigma$, the Helstrom-optimal Bayes error satisfies*

$$P_e^\star(\rho,\sigma) \le \tfrac{1}{2}\, Q(\rho,\sigma). \tag{3}$$

Definition 5 (Observer-effective Chernoff exponent on a fragment set)**.** For an observer O and fragment set A, define

$$\xi_O(A) := \max_{s\in[0,1]} \left(-\sum_{k\in A} \log \mathrm{Tr}\Big[\Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})^s\, \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})^{1-s}\Big] \right). \tag{4}$$

Proposition 1 (Sufficient access condition for δ-decoding)**.** *Fix a target $\delta \in (0,\frac{1}{2})$. If there exists a fragment set A with $|A| \le m_{\max} = \lfloor R_O N \rfloor$ such that*

$$\xi_O(A) \ge \log\left(\frac{1}{2\delta}\right),$$

then O can decode X with error probability at most δ in the SBS witness experiment.

Theorem 2 (Calibration degrades Chernoff distinguishability)**.** *Let Λ be a CPTP map and ρ,σ quantum states. For every $s \in [0,1]$,*

$$\mathrm{Tr}\big[\Lambda(\rho)^s \Lambda(\sigma)^{1-s}\big] \;\ge\; \mathrm{Tr}\big[\rho^s \sigma^{1-s}\big].$$

Equivalently, $\xi(\Lambda(\rho),\Lambda(\sigma)) \le \xi(\rho,\sigma)$.

IV. HETEROGENEOUS FRAGMENTS AND OPTIMAL ACCESS

QD models often assume identical records, but concrete open-system models can produce heterogeneous fragment distinguishabilities (e.g., position-dependent scattering, nonuniform couplings). In the SBS witness experiment, each fragment k induces a per-fragment log-overlap term

$$\ell_k(s) := -\log \mathrm{Tr}\Big[\Lambda_O(\sigma_{\mathcal{E}_k}^{(0)})^s\, \Lambda_O(\sigma_{\mathcal{E}_k}^{(1)})^{1-s}\Big].$$

For a chosen set A, the exponent is $\max_{s\in[0,1]} \sum_{k\in A} \ell_k(s)$. When the minimizing s is common across fragments (e.g., commuting families, many symmetric models), the exponent becomes additive in natural "per-fragment Chernoff informations". This makes the *access* constraint $|A| \le m_{\max}$ a combinatorial resource-allocation problem: which fragments should be sampled to maximize distinguishability.

Proposition 2 (Optimal subset under pure-record additivity)**.** *Suppose the conditional fragment states are pure for each k, so that Q equals the fidelity and $\ell_k(s)$ is s-independent. Then the optimal exponent over all sets A of size m is obtained by choosing the m fragments with the largest per-fragment exponents.*

V. ε-SBS ROBUSTNESS OF DECISION RULES

Lemma 1 (Decision-theoretic continuity under trace distance)**.** *Let ρ and σ be states on the same Hilbert space and let* Dec *be any measurement-plus-decision rule producing a classical hypothesis $\widehat{X}$. Then*

$$\Big|{}_\rho(\widehat{X} \neq X) - {}_\sigma(\widehat{X} \neq X)\Big| \le D_{\mathrm{tr}}(\rho,\sigma).$$

Theorem 3 (ε-robust Chernoff sample complexity)**.** *Let $\rho_{S\mathcal{E}}$ be ε-SBS with witness $\sigma_{S\mathcal{E}}$. Fix a fragment set A and an observer O. If $\delta > \varepsilon$ and*

$$\xi_O(A) \ge \log\left(\frac{1}{2(\delta-\varepsilon)}\right),$$

then there exists a decision rule on A such that ${}_\rho(\widehat{X} \neq X) \le \delta$.

VI. LOCAL VERSUS COLLECTIVE DECODING: A SHARP GAP FOR PURE RECORDS

Temporal horizons can restrict feasible measurements. A key example is the distinction between collective measurements across m copies (requiring quantum memory for coherence) and product measurements (measuring each copy immediately).

For two pure states $|\psi_0\rangle, |\psi_1\rangle$ with overlap $c = |\langle\psi_0|\psi_1\rangle| \in (0,1)$, the m-copy states have overlap c^m and quantum Chernoff coefficient c^{2m} [11]. By contrast, any measurement on a single copy induces classical distributions p,q whose Bhattacharyya coefficient is bounded below by the square-root fidelity [15, 16]. For pure states this yields $B \ge c$, so product measurements cannot beat an exponent of $-\log c$ per copy.

Corollary 1 (A factor-of-two exponent gap for pure-state records)**.** *Let $P_e^{\mathrm{coll}}(m)$ be the optimal Bayes error using collective measurements on m copies, and $P_e^{\mathrm{prod}}(m)$ the optimal Bayes error over all product measurements. Then*

$$\lim_{m\to\infty} -\frac{1}{m}\log P_e^{\mathrm{coll}}(m) = -\log(c^2), \qquad \limsup_{m\to\infty} -\frac{1}{m}\log P_e^{\mathrm{prod}}(m) \le$$

VII. WORKED PHYSICAL MODEL: CENTRAL-SPIN PURE DEPHASING

We provide a concrete open-system model and compute observer parameters from physical couplings and noise.

A. Model and fragment overlap

Let S be a qubit with pointer basis the eigenbasis of σ_z^S, and $\mathcal{E}$ consist of N qubits. Consider

$$H = \sigma_z^S \otimes \sum_{k=1}^{N} g_k \sigma_z^{(k)}, \tag{5}$$

and initial environment state $|+\rangle^{\otimes N}$. Conditioned on the pointer value x (eigenvalues ± 1 of σ_z^S), fragment k evolves to $|\phi_k^{(x)}(t)\rangle = e^{\mp i g_k t \sigma_z} |+\rangle$, so the overlap is

$$c_k(t) = \left| \langle \phi_k^{(0)}(t) \rangle \, \phi_k^{(1)}(t) \right| = |\cos(2 g_k t)| \, . \tag{6}$$

This model is standard in decoherence theory and connects to SBS formation in spin environments [5, 17].

B. Chernoff exponent and redundancy

For pure states, the Chernoff coefficient equals the fidelity, so the per-fragment exponent is $-\log c_k(t)^2$. For a set A of accessed qubits,

$$\xi(A;t) = \sum_{k \in A} -\log\!\big(\cos^2(2 g_k t)\big) \, . \tag{7}$$

A sufficient condition for error $\leq \delta$ is $\xi(A;t) \geq \log(1/(2\delta))$.

C. Mapping to (R_O, C_O, τ_O)

Access. If $m_{\max}$ qubits are interceptable per episode out of N, then $R_O = m_{\max}/N$.

Calibration. Suppose the observer's instrument applies a depolarizing channel to each accessed qubit: $\Lambda_p(\rho) = (1-p)\rho + p\frac{\mathbf{1}}{2}$. Depolarization contracts trace distance by factor $(1-p)$ [14]. For this model, the single-copy Helstrom success probability becomes

$$C_O^{(k)}(t) = \tfrac{1}{2}\left(1 + (1-p)\sqrt{1 - c_k(t)^2}\right) = \tfrac{1}{2}\left(1 + (1-p)|\sin(2 g_k t)|\right)$$

Time horizon. Collective decoding across m fragments requires coherence time at least the acquisition time plus decoding time. If t_{meas} is per-qubit acquisition time, one feasibility constraint is $\tau_O \gtrsim m \, t_{\text{meas}}$. Additionally, pointer stability over the acquisition interval is typically limited by system relaxation times (e.g., T_1) [18, 19].

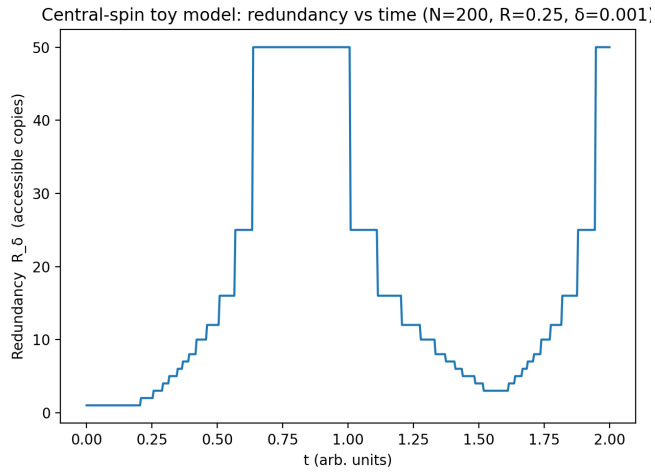

FIG. 1. Central-spin toy model: redundancy (accessible copies) versus interaction time for a sampled coupling distribution, access fraction $R_O = 0.25$, and target error $\delta = 10^{-3}$. Produced by the reproducibility script in the repository.

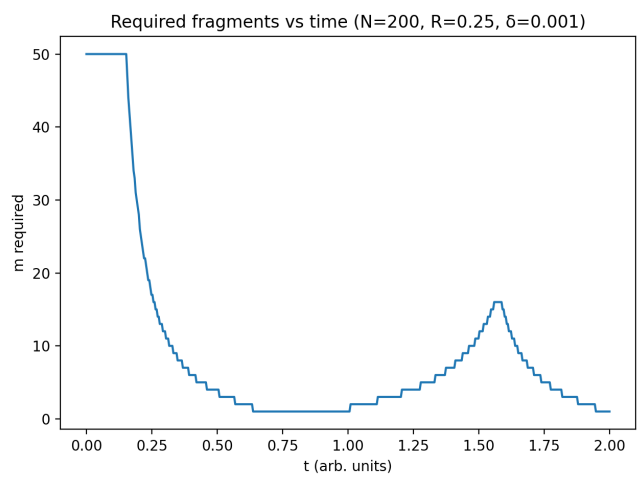

FIG. 2. Same setting as Fig. 1: minimal number of accessed qubits required (under best-subset selection) to achieve target error $\delta = 10^{-3}$ using the Chernoff upper bound.

VIII. CONCLUSION

Explicit observer-quality parameterization yields decoder-level, ε-robust redundancy bounds for QD/SBS. Calibration is a provable contraction of the Chernoff exponent, and temporal limitations can force measurement restrictions with quantifiable performance loss (Cor. 1). The central-spin example shows how observer parameters can be computed from couplings and noise, enabling concrete predictions.

Appendix A: Proof sketches and standard lemmas

1. Proof of Theorem 1

Inequality (3) is a standard consequence of the quantum Chernoff bound [11].

2. Proof of Theorem 2

For $s \in (0,1)$, define the Petz-Rényi divergence $D_s(\rho\|\sigma) = \frac{1}{s-1}\log \mathrm{Tr}[\rho^s\sigma^{1-s}]$. Data processing under CPTP maps holds for $s \in (0,1)$ [20], yielding the claimed inequality.

3. Proof of Lemma 1

A measurement-plus-decision rule is a CPTP map from states to a classical distribution; trace distance contracts under CPTP maps [14], hence error probabilities differ by at most $D_{\mathrm{tr}}(\rho,\sigma)$.

4. Proof of Theorem 3

Apply Proposition 1 to the witness SBS state to obtain error $\leq \delta - \varepsilon$ under σ, then use Lemma 1.

5. Proof of Proposition 2

For pure states, Q equals fidelity and is multiplicative under tensor products, so maximizing $\xi(A;t)$ over A reduces to maximizing an additive sum of per-fragment exponents.

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